

Recovering Deleted and Damaged Files (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 03

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Time on Task:

3 hours, 42 minutes

Progress:

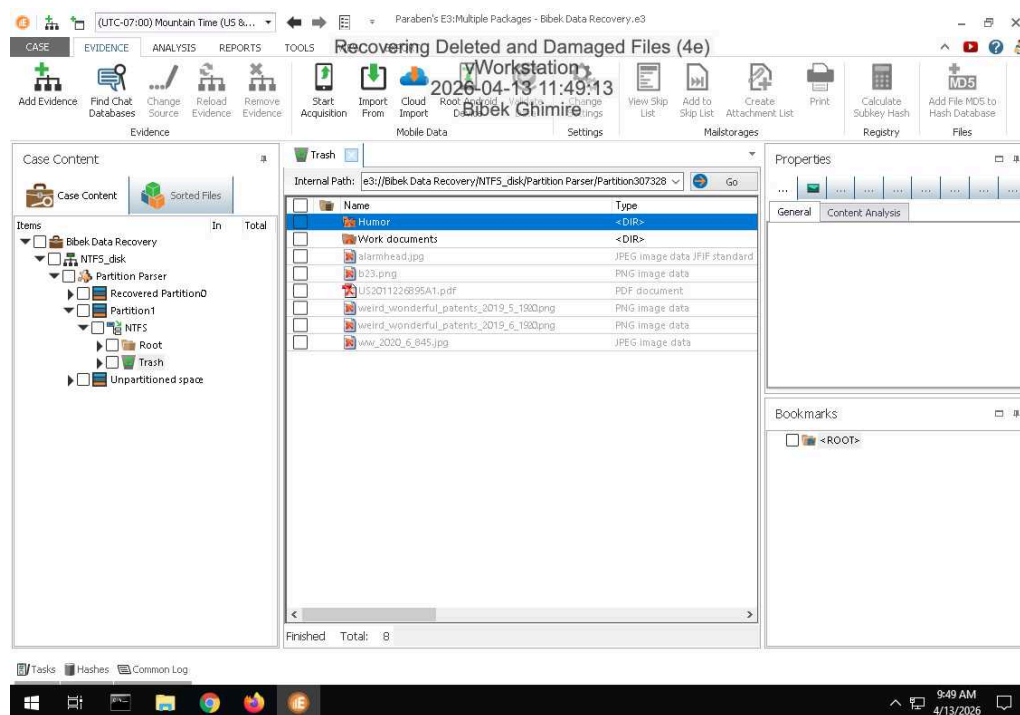
100%

Report Generated: Thursday, June 25, 2026 at 4:40 PM

Section 1: Hands-On Demonstration

Part 1: Recover Deleted Files from an NTFS Drive Image with E3

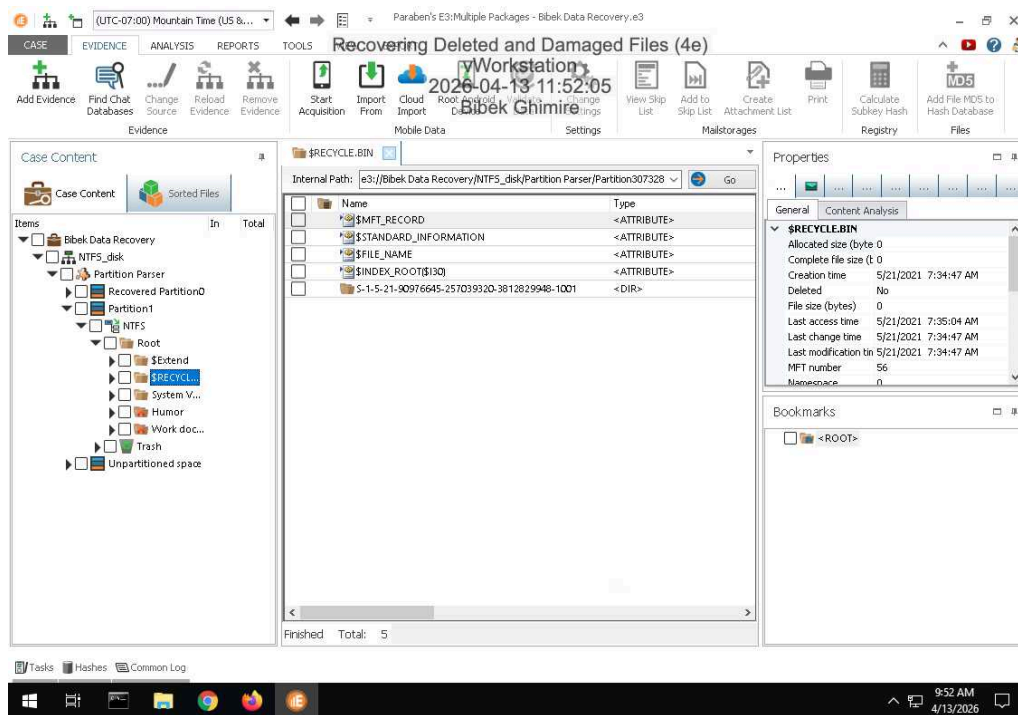
13. Make a screen capture showing the list of recovered files and folders in the E3 Trash folder.



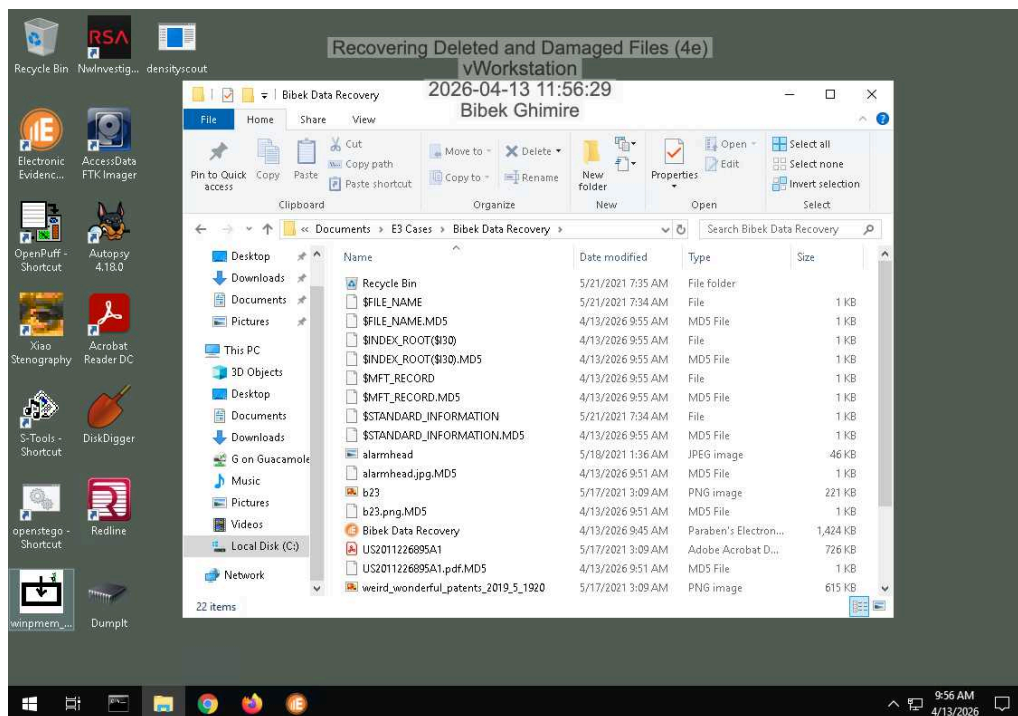
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20. Make a screen capture showing the patent file in the File Viewer.



25. Make a screen capture showing the recovered files in the File Explorer.

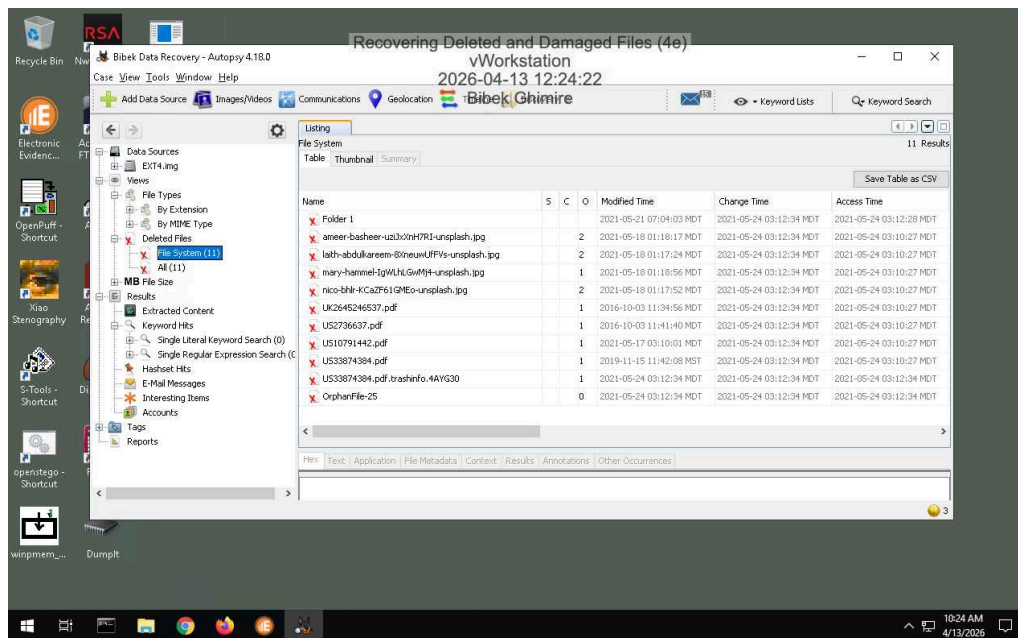


Part 2: Recover Deleted Files from an Ext4 Drive Image with Autopsy

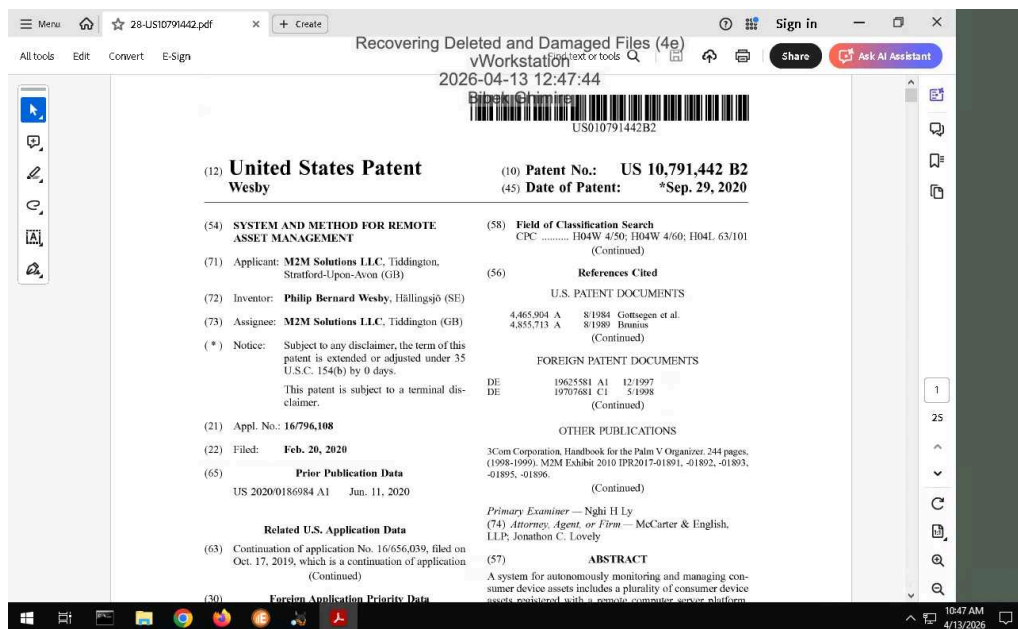
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14. Make a screen capture showing the contents of the list of deleted files in Autopsy.



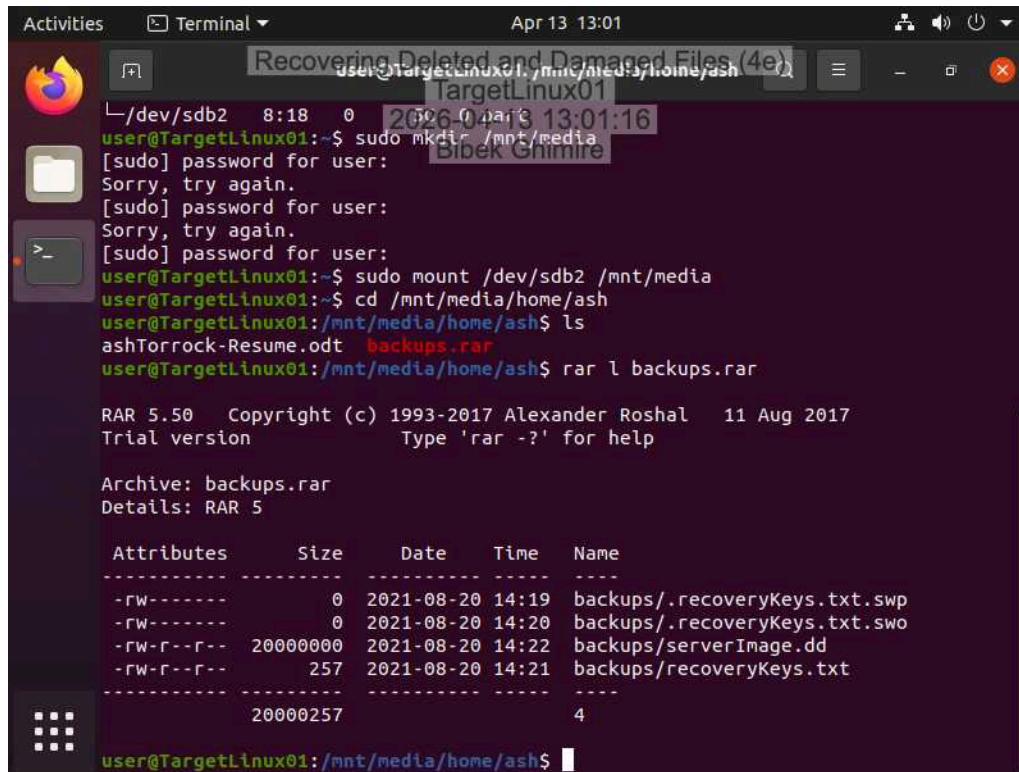
22. Make a screen capture showing the recovered patent file.



Section 2: Applied Learning

Part 1: Recover Deleted Files in Linux with PhotoRec

9. Make a screen capture showing the contents of the RAR archive in the `/mnt/media/home/ash` directory.



A terminal window titled "Recovering Deleted and Damaged Files (4e)" on a system named "TargetLinux01". The user is at the prompt `user@TargetLinux01:~$`. The terminal shows the following sequence of commands and output:

```
user@TargetLinux01:~$ sudo mkdir /mnt/media
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
user@TargetLinux01:~$ cd /mnt/media/home/ash
user@TargetLinux01:/mnt/media/home/ash$ ls
ashTorrock-Resume.odt  backups.rar
user@TargetLinux01:/mnt/media/home/ash$ rar l backups.rar
```

The output of the `rar l backups.rar` command shows the following details:

```
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5
```

Attributes	Size	Date	Time	Name
-rw-----	0	2021-08-20	14:19	backups/.recoveryKeys.txt.swp
-rw-----	0	2021-08-20	14:20	backups/.recoveryKeys.txt.swo
-rw-r--r--	20000000	2021-08-20	14:22	backups/serverImage.dd
-rw-r--r--	257	2021-08-20	14:21	backups/recoveryKeys.txt

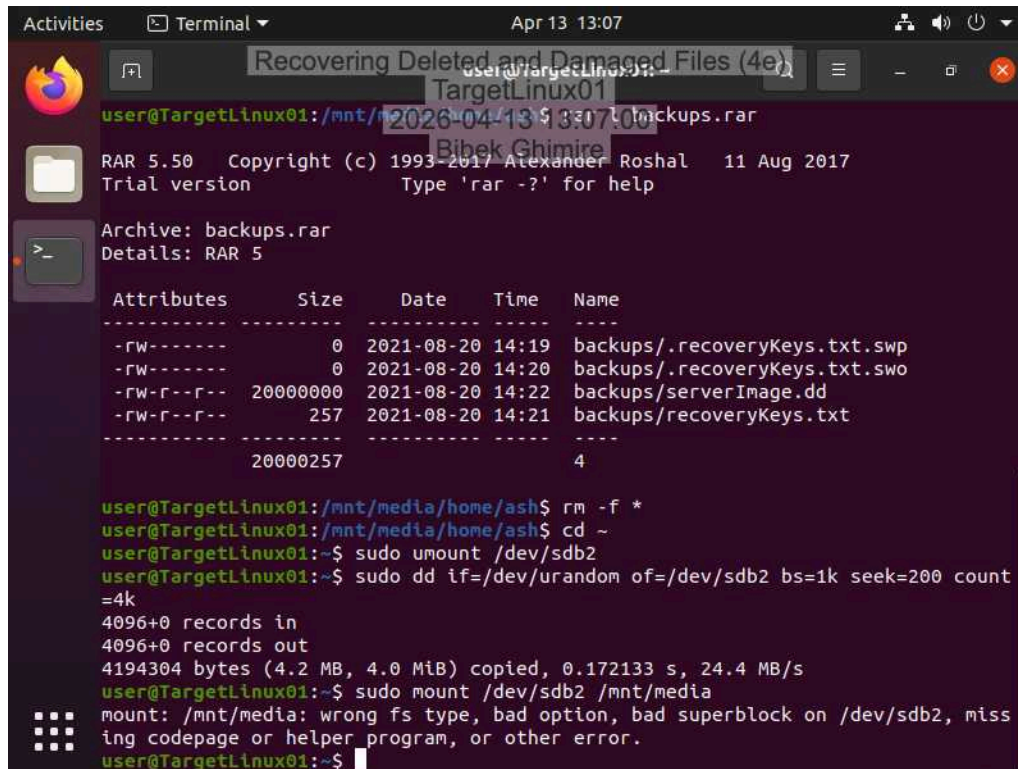
				20000257 4

The terminal ends with the prompt `user@TargetLinux01:/mnt/media/home/ash$`.

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15. Make a screen capture showing the failed mount attempt on the /dev/sdb2 device.



A terminal window titled "Recovering Deleted and Damaged Files (4e)" showing the extraction of a RAR file named "backups.rar". The output lists the contents of the archive, including recovery keys and a server image. After removing the files, an attempt is made to mount the /dev/sdb2 device, which fails with an error message: "mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error."

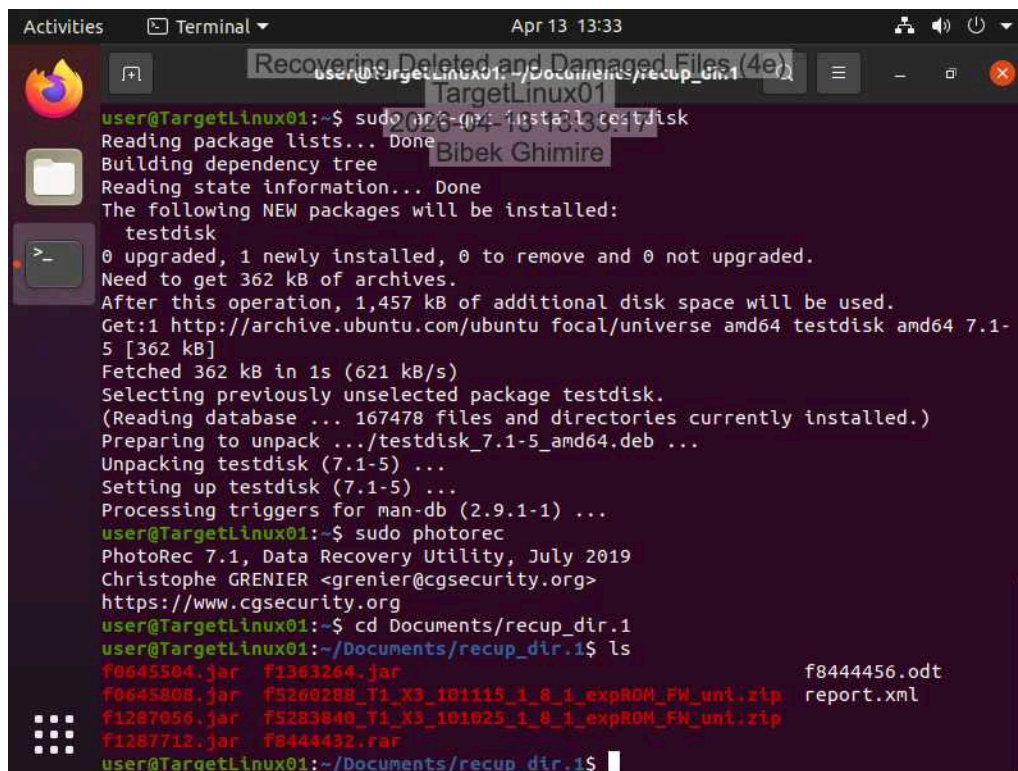
```
user@TargetLinux01:~$ sudo tar -xvf backups.rar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version
Type 'rar -?' for help

Archive: backups.rar
Details: RAR 5

Attributes      Size      Date      Time      Name
-----
-rw-r--r--      0      2021-08-20 14:19 backups/.recoveryKeys.txt.swp
-rw-r--r--      0      2021-08-20 14:20 backups/.recoveryKeys.txt.swo
-rw-r--r-- 20000000 2021-08-20 14:22 backups/serverImage.dd
-rw-r--r--    257      2021-08-20 14:21 backups/recoveryKeys.txt
-----
                20000257              4

user@TargetLinux01:/mnt/media/home/ash$ rm -f *
user@TargetLinux01:/mnt/media/home/ash$ cd ~
user@TargetLinux01:~$ sudo umount /dev/sdb2
user@TargetLinux01:~$ sudo dd if=/dev/urandom of=/dev/sdb2 bs=1k seek=200 count=4k
4096+0 records in
4096+0 records out
4194304 bytes (4.2 MB, 4.0 MiB) copied, 0.172133 s, 24.4 MB/s
user@TargetLinux01:~$ sudo mount /dev/sdb2 /mnt/media
mount: /mnt/media: wrong fs type, bad option, bad superblock on /dev/sdb2, missing codepage or helper program, or other error.
user@TargetLinux01:~$
```

32. Make a screen capture showing the compressed files recovered by PhotoRec.



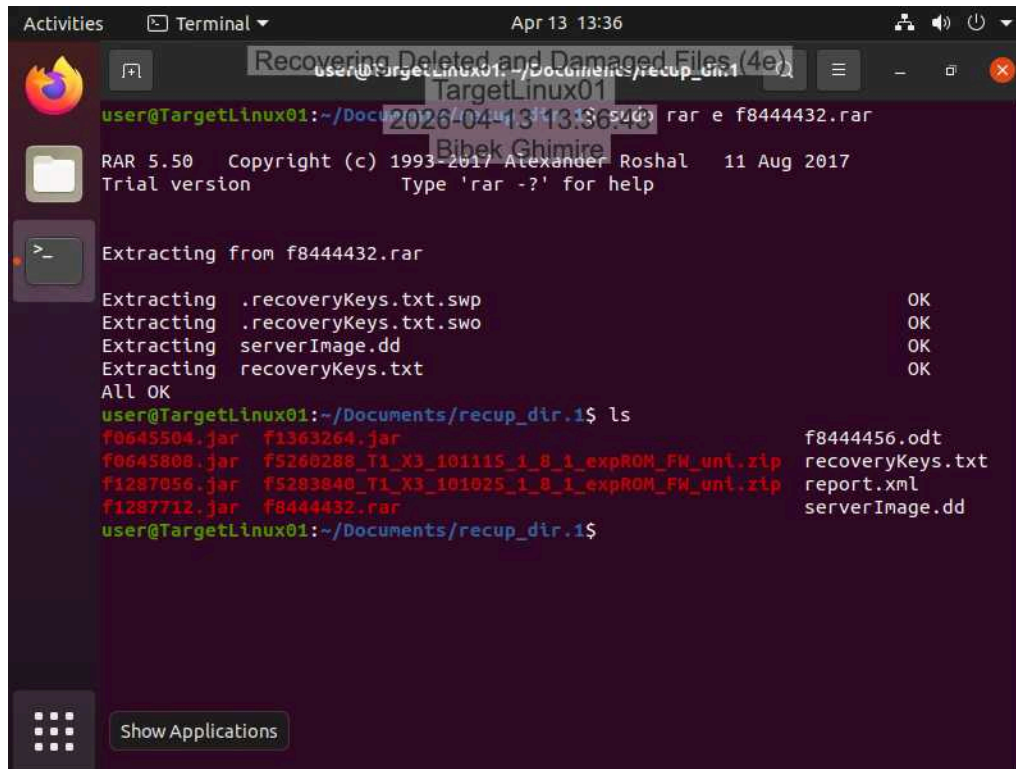
A terminal window titled "Recovering Deleted and Damaged Files (4e)" showing the installation of the testdisk package. After installation, the PhotoRec utility is run, which recovers several files from a disk image. The recovered files are listed at the bottom of the terminal output.

```
user@TargetLinux01:~$ sudo apt-get install testdisk
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following NEW packages will be installed:
testdisk
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 362 kB of archives.
After this operation, 1,457 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu focal/universe amd64 testdisk amd64 7.1-5 [362 kB]
Fetched 362 kB in 1s (621 kB/s)
Selecting previously unselected package testdisk.
(Reading database ... 167478 files and directories currently installed.)
Preparing to unpack .../testdisk_7.1-5_amd64.deb ...
Unpacking testdisk (7.1-5) ...
Setting up testdisk (7.1-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@TargetLinux01:~$ sudo photorec
PhotoRec 7.1, Data Recovery Utility, July 2019
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
user@TargetLinux01:~$ cd Documents/recup_dir.1
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar f1363264.jar f8444456.odt
f0645808.jar f5260288_T1_X3_101115_1_8_1_expROM_FW_unl.zip report.xml
f1287056.jar f5283840_T1_X3_101025_1_8_1_expROM_FW_unl.zip
f1287712.jar f0444432.rar
user@TargetLinux01:~/Documents/recup_dir.1$
```

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35. Make a screen capture showing the backup files recovered from the RAR archive.



A terminal window titled "Recovering Deleted and Damaged Files (4e)" showing the extraction of a RAR archive. The user is at the prompt `user@TargetLinux01:~/Documents/recup_dir.1$` and runs `rar e f8444432.rar`. The terminal output shows the RAR version (5.50), copyright (1993-2017 Alexander Roshal), and trial status. It then lists the files being extracted: `.recoveryKeys.txt.swp`, `.recoveryKeys.txt.swo`, `serverImage.dd`, and `recoveryKeys.txt`, all with "OK" status. Finally, it shows the contents of the directory `~/Documents/recup_dir.1` after extraction, listing several .jar files, .zip files, and other files like `f8444456.odt`, `recoveryKeys.txt`, `report.xml`, and `serverImage.dd`.

```
user@TargetLinux01:~/Documents/recup_dir.1$ rar e f8444432.rar
RAR 5.50 Copyright (c) 1993-2017 Alexander Roshal 11 Aug 2017
Trial version Type 'rar -?' for help

Extracting from f8444432.rar
Extracting .recoveryKeys.txt.swp OK
Extracting .recoveryKeys.txt.swo OK
Extracting serverImage.dd OK
Extracting recoveryKeys.txt OK
All OK
user@TargetLinux01:~/Documents/recup_dir.1$ ls
f0645504.jar f1363264.jar f8444456.odt
f0645808.jar f5260288_T1_X3_101115_1_8_1_expROM_FW_unl.zip recoveryKeys.txt
f1287056.jar f5293840_T1_X3_101025_1_8_1_expROM_FW_unl.zip report.xml
f1287712.jar f8444432.rar serverImage.dd
user@TargetLinux01:~/Documents/recup_dir.1$
```

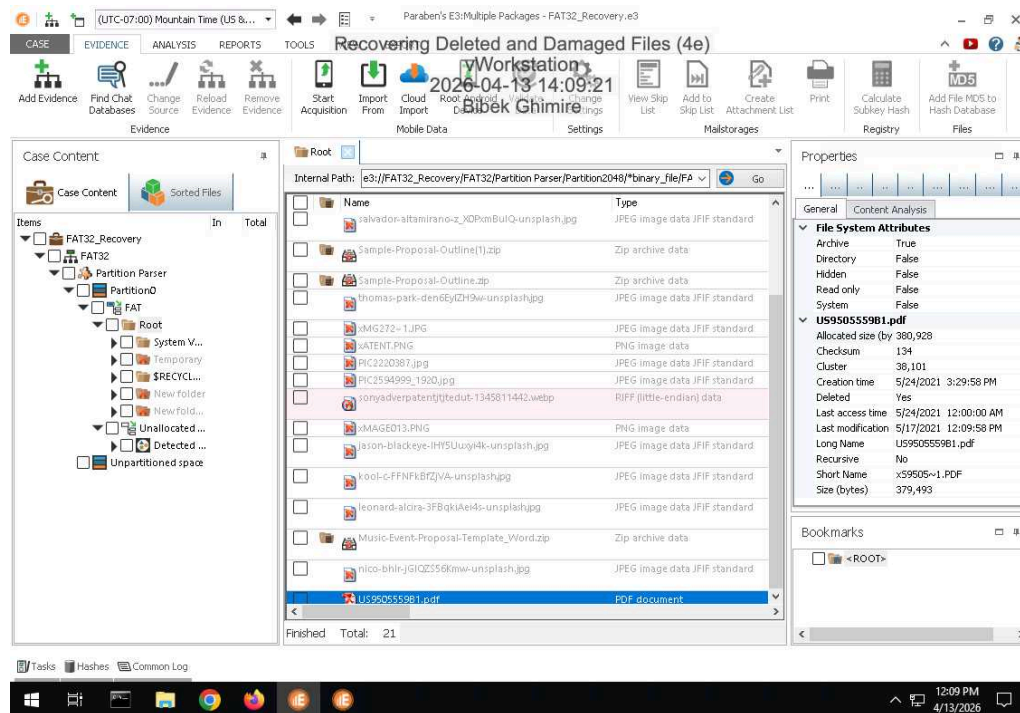
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Section 3: Challenge and Analysis

Part 1: Recover Deleted Files from a FAT Drive Image

Make a screen capture showing the patent file recovered from the FAT32 drive image within E3.



Part 2: Recover Deleted Files from a APFS Drive Image

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Make a screen capture showing the patent file recovered from the APFS drive image within Autopsy.

